Ollama

Ollama
1. Large Language Models (LLMs)
2. Offline Models and 4GB Compatibility
3. Installing Ollama
    Script installation
4. Using Ollama
5. Uninstalling Ollama
References

> **Demo environment**

Development board: Jetson Orin series mainboards

Ollama is an open source tool designed to simplify deploying and running large language models, so that users can run high-quality language models in a local environment.

1. Large Language Models (LLMs)

Large Language Models (LLMs) are a class of advanced text generation systems built on artificial intelligence technology. Their main characteristic is that they can learn and understand human language from large-scale training data and can generate natural and fluent text.

2. Offline Models and 4GB Compatibility

Before downloading a model, first select a version based on the board's memory. The table below summarizes the models covered in this chapter's offline model tutorials and adds the Qwen3-VL model used by the Chapter 11 visual understanding case. Compatibility is assessed for **Jetson Orin Nano 4GB with this course's configuration**; it is not a general conclusion for every quantization, context length, and runtime configuration of these models.

The table marks "Supported" or "Not supported" according to this course's compatibility results; even a supported model still needs memory reserved for the system and other runtime components. The model download size does not equal the runtime memory: inference also requires context cache, vision components, and system memory.

Model family and tutorial	Model version	Main use	Orin Nano 4GB
Llama 3.2	`llama3.2:1b`	Text chat	Supported
Llama 3.2	`llama3.2:3b`	Text chat	Supported
Qwen3	`qwen3:0.6b`	Text chat	Supported
Qwen3	`qwen3:1.7b`	Text chat	Supported
Qwen3	`qwen3:4b`	Text chat	Supported
Qwen3	`qwen3:8b`	Text chat	**Not supported**

Model family and tutorial	Model version	Main use	Orin Nano 4GB
Phi-4-mini	`phi4-mini:3.8b`	Text chat, reasoning	**Not supported**
DeepSeek-R1	`deepseek-r1:1.5b`	Text chat, reasoning	Supported
DeepSeek-R1	`deepseek-r1:7b`	Text chat, reasoning	**Not supported**
DeepSeek-R1	`deepseek-r1:8b`	Text chat, reasoning	**Not supported**
Qwen2.5-VL	`qwen2.5vl:3b`	Image understanding, text chat	**Not supported**
Qwen2.5-VL	`qwen2.5vl:7b`	Image understanding, text chat	**Not supported**
Qwen3-VL	`qwen3-vl:2b`	Image understanding, text chat	**Not supported (this course's multimodal case)**
Gemma 3	`gemma3:1b`	Text chat	Supported
Gemma 3	`gemma3:4b`	Image understanding, text chat	**Not supported**
LLaVA	`llava:7b`	Image understanding, text chat	**Not supported**
MiniCPM-V	`minicpm-v:8b`	Image understanding, text chat	**Not supported**
LocateAnything	LocateAnything-3B Q4_K_M + BF16 mmproj	Visual object grounding	**Not supported**

> **Course scope:** The range of small-parameter text models in the table above does not mean that complete multimodal applications can also run. The six cases in Chapter 11, "Multimodal large model application (online/offline large model + text/voice interaction)", do not support a 4GB configuration; use a Jetson Orin Nano 8GB. The fact that a model itself can run does not mean it supports the image input or tool calling required by the cases.

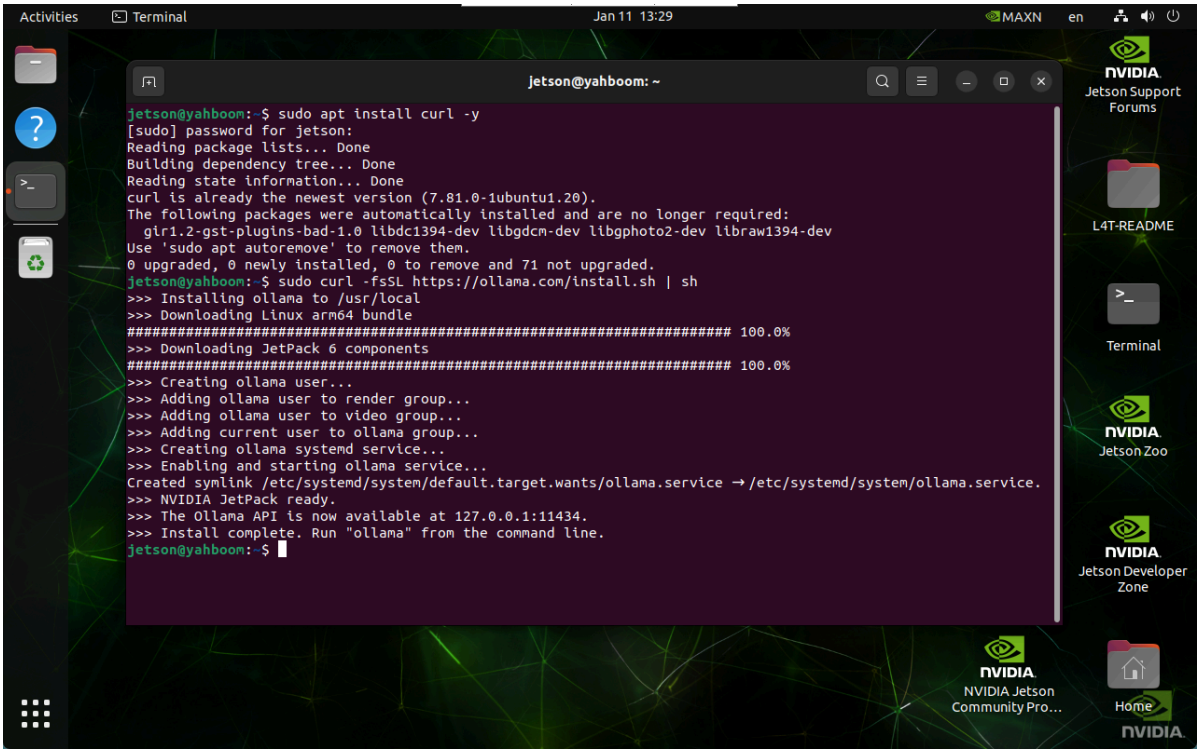
3. Installing Ollama

This tutorial demonstrates installing Ollama with a script on Jetson Orin series mainboards. It is already installed on the factory system, so you can skip this section.

Note: If pulling from Ollama fails, that is normal. Try several times, or pull through a proxy. How to set up a proxy is something you need to look up yourself.

Script installation

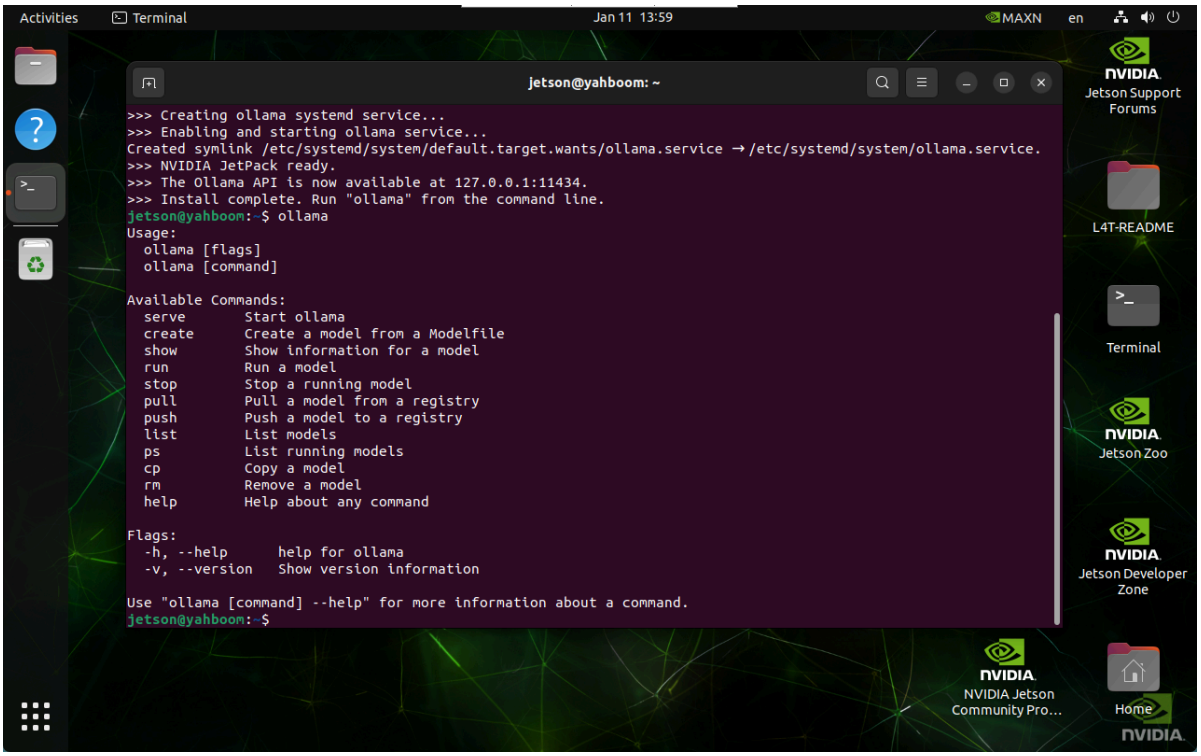
```
sudo apt install curl -y
sudo curl -fsSL https://ollama.com/install.sh | sh
```



The whole installation process takes a long time; please be patient.

4. Using Ollama

Enter `ollama` in the terminal to see the help output:



Command	Purpose
ollama serve	Start Ollama
ollama create	Create a model from a model file
ollama show	Show model information
ollama run	Run a model
ollama pull	Pull a model from the registry
ollama push	Push a model to the registry
ollama list	List models
ollama ps	List running models
ollama cp	Copy a model
ollama rm	Remove a model
ollama help	Get help on any command

5. Uninstalling Ollama

- Remove the service

```
sudo systemctl stop ollama
sudo systemctl disable ollama
sudo rm /etc/systemd/system/ollama.service
```

- Remove the files

```
sudo rm $(which ollama)
```

- Remove the models, service user, and group

```
sudo rm -r /usr/share/ollama
sudo userdel ollama
sudo groupdel ollama
```

References

> **Ollama**

Official website: https://ollama.com/

GitHub: https://github.com/ollama/ollama